

SIERRA WILDLIFE RESCUE SQUIRREL MANUAL



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SQUIRREL SPECIES OVERVIEW

Squirrels are one of the most ancient of all mammal species. The oldest squirrel fossil ever found dates back 55 million years and there are over 280 species, in almost all geographic regions of the world. El Dorado County, alone, has seven species of squirrels, though SWR usually receives only six for rehabbing. The seventh is the Golden-Mantled Ground Squirrel, found near South Lake Tahoe and higher elevations.

Of all the species found in California, the Western Gray Squirrel tree squirrel is the most abundant. It is easily recognized by its large size (340 - 964 grams, or 1.5 - 2 lbs.), its tuxedo-like marking of a silver-gray back and legs, with white on its underside, and white rings around its eyes, and its beautiful, long, bushy tail, often curving over its back like an umbrella. The genus for squirrels, *Sciurus*, in fact, comes from the Greek words *skia*, for "shadow," and *oura*, "tail," since many tree squirrels have this feature. The Western Gray's color phases also include black (melanistic) and, rarely, pure white.

The Fox Squirrel originated in the East, but was introduced in the West in 1904. Although normally considered the largest of the tree squirrels in the East, in the West, the Fox Squirrel is considerably smaller than the Western Gray. Its coloring is grayish-brown on its back and head, with tan to rust markings on its face, legs, feet and underparts. The underside of its tail may range from tan or rust to a brilliant orange. It may occasionally have white markings on its face, the tip of its tail, or feet. Fox squirrels are more frequently melanistic than grays.

The Douglas Squirrel, also known as the Chickaree or Pine Squirrel, is a small (250 - 300 grams, or 5.25 oz.), very vocal, energetic squirrel about ¼ the size of the larger tree squirrels, with gray-to-brown fur, a dark brown stripe down each side, beige markings on its face and underside, and a medium-length, flat tail. It is found most often at elevations from 2,500 to 6,000 ft.

The California Ground Squirrel is also abundant, and usually a little smaller than the Western Gray (454 - 625 grams, or 1 lb - 1 lb. 6 oz.), with stippled fur of gray and light brown over the head and back, and a V-shaped, whitish mantle over its shoulders; it has white-ringed eyes, buff-colored underparts, and a shorter, straighter bushy tail than the Gray. It lives in large colonies, mainly in ground burrows as long as 200 feet and 30 feet deep, on hillsides or in rocky areas.

The Northern Flying Squirrel is a very small squirrel (75 – 150 grams, or 1.5 - 5 oz.), seen rather infrequently in the wild. It has silky gray or cinnamon-brown fur, a wide, feathery tail, and a furred, fleshy membrane (*patagium*) that extends from the wrist of the foreleg to the ankles, enabling it to glide for long distances between tree branches (up to 300 feet from the highest branches of tall trees). It has enormous eyes, for night vision, since it is the only nocturnal squirrel.

Chipmunks are also members of the squirrel family, and several species exist throughout California. They range from 2.5 – 5 oz at maturity, and are light to dark brown, with a variety of striping patterns of black, and white. They are mainly terrestrial, although they can also climb trees. They prefer mountainous regions, and nest in burrows, tree cavities, logs, and stumps.

GENERAL INFORMATION: It is very important that any baby squirrel be raised with its own kind. Many of the skills necessary for survival are learned in play and other interactions with other squirrels. Not only do we want to encourage normal behavioral development but, as it matures, we want to minimize the animal's dependence on humans. Since most tree squirrels have litters ranging from 2-4, rescued babies are often found in pairs or even trios; if they are not, we encourage rehabbers who acquire just one baby squirrel to pair it up with another single baby of similar age as soon as possible. Weaned juveniles or squirrels more than two weeks apart in age should not generally be paired up, nor should squirrels of different species. If unsure, experienced rehabbers can be consulted before pairing up squirrels.

The first thing to do when you acquire a squirrel is to move him into a ventilated basket, with a snap-shut lid, or a well-ventilated box, well-padded with soft clothes, such as baby blankets, t-shirt or soft sweat-shirt material. Place him in a warm, dark and quiet location, without any household traffic, on a heating pad, set on Low. Be sure there is thick padding under the squirrel, and place some of the material over him lightly. You can perform a **quick** examination to determine if he has life-threatening conditions (e.g. profuse bleeding, breathing or broken bone problems, or severe shock); if so, the appropriate treatment should be given immediately.

It is important to determine the squirrel's weight as soon as he is warmed up. You can weigh him in a small, lightweight basket or plastic tub, placing him first in a small cloth bag, or even a clean cotton sock, depending on his size. Weigh the container first, then tare the container (or deduct its weight) and weigh with the

squirrel. It is very important to check the squirrel's weight daily, ideally at the same feeding time, for several reasons, including 1) determining his approximate age; 2) determining the amount of fluids or formula he should have; and, 3) determining the correct amount of medicine he needs for any illness or condition. Rapidly losing or gaining weight is also a significant gauge of his health. You will also need to know his weight before you put him into a Release Cage for eventual release to the wild.

Since small animals are usually weighed in grams (g), having a high-quality digital gram scale (0.5 – 1000 grams) is preferred, but you can also weigh him on a good meat/produce or postage scale and convert ounces to grams (1 oz = 28 grams). (See *Useful Measurement Conversions* chart, Attachment 2). Healthy weight gains per day average about the following: Gray squirrels should gain from 2-3 grams per day at the age of 2-3 weeks, 4-7 grams at 4-6 weeks, and about 8-10 grams per day when older; the smaller squirrel species gain about 2-4 grams per day.

It is also important to keep accurate records each time you feed, medicate, or carry out any other procedure with your squirrel, as these are required by SWR and DFW. They are also useful to any other rehabber to whom you may transfer the squirrel for any reason. You can fill out the Intake/Medical Record for each animal or keep records in WRMD on-line. If you are the initial rehabber, you will also be assigned a number for each squirrel by the Team Leader, as required by DFW. If using paper records, as you add pages to the record, you will transfer the initial information – squirrel's number, date taken in, species and sex of squirrel, etc. to each of the additional pages.

NOTE: Any squirrel will need warming and hydration before giving formula. It is assumed that a novice rehabber will be getting only a healthy baby squirrel(s) from his/her mentor, and these steps will have already been taken. For information on initial intake, including hydration, see Section 1.2.

1. FORMULAS AND FEEDING

Experienced squirrel rehabbers have tested different types of formula over many years and have found that Fox Valley Day 1 Squirrel Formula is the most nutritionally complete and closest to the mother squirrel's milk. Fox Valley squirrel formulas come in two types: 32/40 for squirrel babies under 4 weeks and 20/50 for babies 4 weeks and older. SWR stocks both formulas, and Ultraboost, a supplement. The basic, full formula for both types is: 2 parts bottled water (not distilled) to 1 part Fox Valley (2:1 ratio). Use a tablespoon to measure the powder and water, or a measuring cup for the water when making larger amounts of the formula, as your squirrel(s) grow (e.g. $\frac{1}{2}$ cup water (or 8 T) to 4 T powder), keeping the ratio of powder to water constant.

Baby squirrels also need beneficial bacteria in their formula to help them digest it and for immune system support. Lactobacillus acidophilus (LA 200) and L-Glutamine should be added to the formula mix at the ratio of 1/16 t each per 1/4 cup for the 2:1 ratio of the formula.

Examples: 4 T water + 2 T powder + 1/16 t LA 200 + 1/16 t L-Glutamine

8T water + 4 T powder + 1/8 t LA 200 + 1/8 t L-Glutamine

Be sure to increase the amounts of LA 200 and L-Glutamine as you make up larger amount of formula.

1.1 FORMULA PREPARATION

The following instructions should be followed when mixing and preparing the formula:

Use Fox Valley 32/40 for squirrels up to 4 weeks of age; Use Fox Valley 20/50 for squirrels over 4 weeks of age.

1. Feed 5%- 7% of body weight at each feeding. Feed amount and number of times per day specified in the *Squirrel Feeding Schedule* depending on age, weight and species.

2. Use 32/40 or 20/50, depending on age.

3. A small jar with a tight lid is handy for mixing formula. If you shake jar or use a whip to mix formula thoroughly, prepare several hours in advance and refrigerate to allow air bubbles to dissipate. Ensure that formula is well-mixed and that all air bubbles are out before you feed. Liquid can also be stirred gently then poured through a sieve to press any clumps through. This method ensures there are no bubbles that could upset squirrel's stomach and can be used right away.

4. For new squirrels, after hydration with LRS oral or SubQ) or Electro Stat (oral only), add amount of water gradually to formula over a couple of feedings. If you are the initial receiver of a squirrel, please read Section 1.2 for complete details on determining hydration needs and feeding schedules at initial intake.

5. Add LA 200 and L-Glutamine to either formula from the beginning for extra intestinal benefit. Another nutritional supplement is *Bioplasma*, which contains essential "cell salts" in an easy-to-use form, which improves the health of a sick or injured squirrel; this can be administered separately at feeding time. We also sometimes use DMG, which boosts the immune system.

6. Beginning at 4-5 weeks of age, switch gradually to 20/50. Blend with the 32/40 your squirrel is used to or feed 20/50 diluted at 3:1 for a couple of feedings up to 24 hours; gradually move to full strength 2 parts water to 1 part powder.

7. Keep formula, refrigerated, for 24 hours, only; make up a new batch daily. Warm only the amount needed for one feeding, in the syringes, in very hot water. **Never microwave** the formula itself; microwaving destroys the nutritional value. A coffee cup or candle warmer is a useful way to keep your water at the right temperature, but some people just microwave it every time.

8. Always store powdered Fox Valley and all supplements in the freezer. Watch expiration dates, and change to new powders when needed. Fox Valley has a very long shelf-life when stored in the freezer.

Handling Formula: For small babies up to 3-4 weeks of age, you should make up just enough formula for one or two feedings at a time, particularly if the baby is sick or not feeding well; the fresher the formula is, the less chance of having introduced any harmful bacteria.

Next, fill a jar or cup with very hot (steaming) tap water or you can microwave the water. Stand the syringe in the water bath until the formula is just warm (not hot), about room temperature, 70 – 72 degrees F; test on the inside of your wrist, like a baby's bottle. Feed immediately, while the formula is still warm; reheat if necessary.

After feeding, dispose of any formula left in the syringes and wash your feeding equipment thoroughly, using hot water and a small bottle brush. Sterilize your equipment every few days by boiling it for 2 minutes, or soak in a good disinfectant, such as Nolvasan, and then rinse it thoroughly with hot water.

Feeding Equipment and Preparation: The formula should be given using only a Luer-Lok, centered-tip, O-ring feeding syringe (1, 3, 5 or 10 cc, depending on the age

of the squirrel), with a Catac, Zoologic or Miracle nipple for the smallest (pinky) babies and a 4-Paws nipple from about 2 weeks on. (All sizes of syringes and nipples are available through SWR). Always use a syringe; a PetNurser bottle will deliver too much air along with the formula, and the squirrel is much more likely to aspirate or become bloated (See *Section 2*).

If the nipple doesn't already have a hole, make a very small hole by snipping off the tip. It may help to heat and run a size 18 needle it in and out several times to enlarge the hole. It must be a **very small hole**. This size hole generally works for babies from birth until 8 weeks. Older babies may be able to use a slightly wider opening. Alternatively, you can slit the tip of the nipple with a razor blade or a sharp or Exacto knife; start with 1 slit; for an older baby, make an "X" slit.

Draw the formula up in the feeding syringe to the correct measurement line on the syringe. If bubbles form, flick the syringe with your finger or tap it on the side of the sink to remove them. You want to get the complete amount of nutritious formula, not air, into your squirrel. Syringe caps work well to heat the formula in the syringe while keeping the water from seeping in. After warming remove cap and put the nipple on the syringe.

1.2 INTAKE PROCEDURES FOR NEWLY- RESCUED SQUIRRELS

Step 1: All rescued squirrels have had a traumatic experience, and should be assumed to be in at least mild shock. When a squirrel is in shock, his temperature will drop. Squirrels are also very susceptible to pneumonia and other respiratory problems. Therefore, all newly-rescued squirrels, no matter what their age or the ambient temperature, should be kept warm and dry. Never start the squirrel on formula until he is well-hydrated with an electrolyte solution.

Do not attempt to feed a “cold” squirrel. Place the squirrel in a basket with a snap lid or a secure, well-ventilated box, lined warmly with several thick layers of cloths, and cover him with other layers. (fleece or flannel baby blankets work well; T-shirt or sweatshirt material also works; **do not** use terry cloth, since little nails can get caught in it.) Make certain he cannot open the cover if he suddenly revives. Place the container in a dark, quiet place away from any noise or traffic, on an electric heating pad set on *Low*. Wrap the animal warmly while feeding. A larger blanket or towel should be placed over the top and sides of the basket to prevent drafts and give him added security. Squirrels should feel “toasty warm” to the touch.

He will be on the heating pad continuously if he is 6 weeks or younger, still in shock, injured, or ill, or if the room is cold. Feel the place in the basket where the squirrel is lying to make sure it is just “toasty-warm,” and check your heating pad often to see if it’s working properly. **After he has warmed up, you can move the basket half-on/half-off the heating pad, so he can move to a cooler spot if he gets too warm.** A room thermometer can be placed near the basket to check room temperature, which should be 72 – 75 degrees.

Homeopathic remedies have been successfully used with squirrels to reduce fear and shock, to calm them down when they first come in, and for other problems. The most widely used of these are *Arnica* (good for any injuries, bruising, etc.) and *Aconite* (for fear and stress). “Rescue Remedy,” available from several sources, is a mixed homeopathic remedy that has also been used successfully for calming a frightened squirrel; spray it on his blankets, not directly on the squirrel. These can be used immediately upon intake, and are easy to administer. Homeopathic remedies, if used properly, cannot injure your squirrel.

Step 2: Weighing: Once your squirrel is warm and calmed down, you should weigh him. As noted earlier, his weight will determine his approximate age, the

amount you should feed him, the status of his health, and give you a basis for many other decisions about how to care for him.

Step 3: Fluid Therapy: Hydrating fluids are given prior to any formula feedings, for two reasons: dehydration and shock. It's safest to assume that all squirrels rescued are somewhat dehydrated and a bit in shock from the trauma of falling, injury or other reasons. You can check for dehydration by carefully pinching the skin between the shoulder blades. If it does not rebound within 1-2 seconds, the squirrel is dehydrated. Other signs of dehydration are "sunken" eyes and dry mucous membranes, and/or no urination or bowel movements or very small, hard dark droppings.

Second, any squirrel rescued will be experiencing a certain level of stress and shock, for which fluids and heat are life-savers. For the first 6-12 hours, (depending on age and circumstances), only Lactated Ringers Solution (LRS) or Electrostat, both hydrating solutions, should be given to a dehydrated squirrel. LRS can be given either orally or subcutaneously; Electrostat can be given **orally only**. However, if a squirrel is injured or ill, he may not readily take liquid orally, and he must be given LRS subcutaneously (SubQ). Giving fluid subcutaneously is the quickest way to hydrate your squirrel, but more invasive than orally. Again, **it is very important not to give formula or any solid food to a dehydrated or cold squirrel, and only LRS or Electrostat to a warmed squirrel at the beginning**

In rehydrating a squirrel, it is important to give the correct amount of Ringer's or Electrostat per day; a good rule of thumb to follow is to give the squirrel 7-8 percent (in severely dehydrated cases, 10% may be given) of his body weight in fluids just for the first couple of feedings, then 5%.

Step 4: Once the squirrel is thoroughly hydrated, young squirrels will get the Fox Valley formula introduced gradually, increasing the strength as directed if there is no sign of soft stools or diarrhea. Guidelines for feeding amounts and schedules for squirrels, by age, are given in *Section 3* and *Squirrel Feeding Schedule*. Start with these first, but once you get to know your squirrel, you may find he may want a little more, or less, at feedings, or to be fed more or less often.

If he is weaned or an adult, after hydration you can try giving him another liquid food to begin with. We primarily use Ensure liquid (you can also mix in some berry yogurt or applesauce, in a blender, to give it a more pleasant flavor), gradually working up to solids if he can eat on his own.

The following are guidelines to follow for introducing formula, following rehydration and stabilization. These are only guidelines; the time until you get the squirrel on full-strength formula will vary, depending on physical condition and responsiveness to feedings. If you have any doubts about how much to hydrate, use the Conservative Schedule.

Newly-Rescued Squirrel Feeding

If **dehydration is mild**, the squirrel may be put on oral LRS or another electrolyte solution for the first 6-12 hours (i.e., if the squirrel hydrates quickly, and is in pretty good condition, it may be placed on an **accelerated schedule** to get it up to full strength formula). If the squirrel is **moderately to severely dehydrated** (and possibly emaciated), the slower, more **conservative schedule** should be used.

ACCELERATED SCHEDULE

First 6-12 hours: Feed with Lactated Ringer's (LRS) or Electro Stat every 2.5 – 3 hours.

Second 6-12 hours and First Week: First 2 feedings: Always use high-quality bottled water (not distilled water, which contains no minerals) to mix with formula. Feed formula made with 4 parts bottled water, 1 part Fox Valley powder along with LA 200 and L-Glutamine. You may mix the formula using Electrostat fluid instead of bottled water for a few feedings, if you're concerned the squirrel is still a little dehydrated.

Next several feedings: Feed formula made with 3 parts water (or Electrostat) to 1 part formula plus supplements.

Second week or when over 60 grams: Feed formula made with 2 parts water to 1 part Fox Valley powder, plus supplements.

CONSERVATIVE SCHEDULE

First 12 – 24 hours: Feed with LRS or Electro Stat, or SubQ injections of LRS (if unable to take orally) every 2.5 – 3 hours. If doing well, change to formula using 4 parts water or Electrostat to 1 part Fox Valley powder during last 12 hours.

Second 2 days: Feed formula made with 3.5 parts water or Electrostat to 1 part Fox Valley powder, plus supplements.

Rest of first week (if well-hydrated): Feed formula made with 3 parts water to 1 part Fox Valley powder, plus supplements. After first week mix at full

formula ratio, 2 parts water to 1 part powder, plus supplements. Feed at this ratio for duration of weeks or months squirrel is on formula, including while weaning from formula to solids.

1.3 FEEDING PROCEDURE

Wrap your squirrel in a baby blanket or soft cloth (leaving the head and front paws free), to keep him warm and decrease stress. You can even cover his head entirely, as long as you can see his mouth and that he's swallowing. Although many rehabbers do not use them at this stage, you may choose to use surgical or cotton gardening gloves to hold your baby squirrel from the beginning; squirrels can carry fleas and other parasites, and some people may feel more comfortable using gloves. Hold the squirrel in your lap, also covered with a towel, thick baby blanket, or soft pillow. Adding several pieces of Kleenex under the squirrel is also helpful, since he may urinate and defecate copiously before or after feeding. Have cotton balls and extra Kleenex handy.

Baby squirrels nurse while lying on their stomachs. Holding the baby on its tummy and gently but firmly around the back of the neck (**not around his throat**), making a sort of "collar" of your thumb and forefinger, with your hand over his upper body. Holding the syringe between your first and second fingers with the heel of your hand on the plunger, direct the nipple into his mouth, tilted slightly upwards. Some people find it easier to use their thumb on the plunger, but there isn't as much control of how fast you are feeding. If the squirrel is feeding too fast, you can use your thumb as a brake on the plunger. **NEVER FEED A SQUIRREL LYING ON ITS BACK;** this will cause aspiration of the formula, which can result in pneumonia and death.

Your rescued baby squirrel will need to get accustomed to the nipple, since it's unlike Mama's, so go very slowly and BE PATIENT. Apply gentle pressure on the syringe plunger with the heel of your hand, in a "push, pause; push, pause" pattern, releasing just a little (about .05 cc) of the formula at a time. To get him started, it sometimes helps to jiggle the nipple a little in his mouth and to squeeze a drop out on the tip so he can smell and taste it. **NEVER PUSH HARD** on the plunger or the baby can aspirate. If he squirms, pulls away or pushes the nipple away with his paws, stop and wait until he settles down, then try again. In the meantime, test the formula to see if it's too hot or cold. For squirrel babies who are reluctant to feed, you can apply a little Karo syrup to the tip of the nipple to start. Try to get the correct amount of formula down at each feeding.

Once he is used to the nipple, he will start kneading with his paws and, eventually, will hold the syringe with his paws and suck hard on the nipple; you may have to control the plunger to keep him from eating too fast. Most baby

squirrels will close their eyes while sucking, but be sure he is awake and swallowing while taking formula, or he may choke.

If he refuses the nipple altogether for several feedings, he may be sick and you need to take immediate steps to determine the reason. Trying different types of nipples may be helpful. After an initial reluctance to use the nipple, most squirrels will let you know when they have had enough, though some will willingly overeat, and you must check that their stomachs are not over-full. The stomach should appear full (just nicely-rounded), but not taut or bloated. When the squirrel is a baby, never feed more than the correct amount, by weight. Overfeeding can cause diarrhea or bloat and can be life-threatening.

Measure the formula in the syringe before and after feeding to determine intake, and record the amount of each feeding on your intake/medical record (Squirrel Intake/Patient Record). Clean any spilled formula off of the squirrel using a cotton ball or tissue, and dry their mouths and penis or other areas if they've urinated or defecated. Make sure you get it all off, even if you have to use a wet washcloth; spilled formula can cause skin abrasions and fur loss. Discard any formula still left in the feeding syringe; do not reuse it.

Stimulation: A baby squirrel needs stimulation to eliminate, and it is very important that he do so. After feeding your squirrel, using a warm, moist or dry cotton ball, gently stroke the genital area to stimulate urination and defecation. For pinkies, a Q-tip sometimes works best for stimulating. It may take from a few seconds up to a couple of minutes or more to get your squirrel to eliminate; be patient. Most small babies will urinate after each feeding and defecate several times per day. Some squirrels need to be stimulated both before and after each feeding. When he urinates, your squirrel should be eliminating about 2/3 of the amount of fluid you just gave him. Wipe any excretions off of the squirrel with a Kleenex or cotton ball and dry him well.

Weigh your squirrel before or after a feeding at the same time every day, until he is at least 8 weeks old, and record his weight in your Intake/Medical Record. This will ensure that you know how much formula he should be getting, and whether he is gaining weight. After 8 weeks, you can weigh your squirrel once or twice a week, unless he is ill, when you should weigh him daily. The larger tree squirrels should gain an average of 5-6 grams per day at the age of 3-5 weeks, and about 8-12 grams per day when older; the smaller squirrel species gain 2-4 grams per day, up to 6

grams when older. Return your squirrel to his basket and cover him up warmly. Until he is about 4 weeks old, he will probably fall asleep immediately after feeding.

2. ILLNESSES, INJURIES AND OTHER PROBLEMS

There are many conditions and problems common to wild animals, as well as those that are somewhat unique to squirrels. Following is a discussion of a few with which the rehabber should become acquainted. Additional problems and illnesses of squirrels are more thoroughly addressed on-line at various rehabilitation sites, from other experienced rehabbers, and are also familiar to your wildlife veterinarian. Although new rehabbers will gain experience in treating many squirrel injuries and illnesses on their own, several veterinarians in El Dorado County also work closely with SWR rehabbers, *pro bono* or at lowered costs, paid through SWR. A listing of veterinarians that will assist in wildlife care is available from SWR.

REMEMBER, DISCUSSION WITH OTHER REHABBERS IS A VERY EFFECTIVE WAY TO RESOLVE MANY PROBLEMS. OTHER REHABBERS MAY HAVE EXPERIENCE WITH A PARTICULAR ISSUE OR MAY SIMPLY PROVIDE A WAY TO TALK THROUGH THE PROBLEM TO RESOLUTION. NEVER HESITATE TO CONSULT WITH YOUR MENTOR OR ANOTHER REHABBER.

BREATHING/RESPIRATION ISSUES:

ASPIRATION

Squirrel babies sometimes aspirate when feeding, getting a nose and/or mouth-full of formula which can be drawn into their lungs.

Feeding techniques to avoid aspiration:

- Never feed a squirrel on its back; this is extremely likely to cause aspiration. Hold him on his stomach and around the shoulders or sitting up, leaning forward (which some squirrels prefer when older).
- Be particularly cautious when you first start (aspiration is most likely to occur when the animal begins feeding and can act ravenously).
- Ensure that the syringe isn't sticking (although this rarely happens with the O-ring type syringe), that the nipple hole isn't too big (delivering the formula too fast for the squirrel), and that the nipple or syringe isn't leaking.

- If aspiration continues to be a problem, be sure to review and correct feeding procedures (i.e. feeding position, using a smaller syringe, feeding more slowly, using the “push-pause, push-pause” pattern, etc.), since correcting one of these will usually eliminate potential aspiration problems.

If he does aspirate:

- Stop feeding, wipe the formula from his nose with a Kleenex, and watch and listen to be certain that he is breathing normally (if he is breathing very fast, too slowly, or from his abdomen, this is abnormal).
- Check his breathing several times after aspiration while feeding (about every three minutes).
- If there is any doubt, use a small bulb syringe; squeezing it just halfway, place it in one nostril at a time, and gently release it to draw the fluid into the syringe. You may have to repeat this several times.
- If any severe breathing problems develop, such as gasping for air, open-mouthed breathing, or gurgling or choking sounds, you will need to act quickly. Check for any blockage in his mouth or throat and remove it if possible. Sometimes holding the squirrel upside down and gently patting his upper back also helps to alleviate breathing problems.
- The best way to check for suspected breathing problems, other than observing the obvious symptoms cited, is to hold the squirrel’s upper back close against your ear; if “crackling” or “clicking” sounds are heard, there is a possibility of pneumonia. A small stethoscope is also a useful tool for assessing breathing. If pneumonia is diagnosed, the best treatment is Nuflor (See Medication Chart for more information.)

BORDATELLA

The Bordatella bacterium has recently appeared in tree squirrels (this is the same bacteria that causes "kennel cough" in your dog). If caught early, it is treatable with TMS and supportive care. If not caught early, it can lead to sudden death. Symptoms can include lethargy, respiratory symptoms, profuse, frequent urination, and sudden weight loss, all or none of which may be present at onset.

PNEUMONIA

Pneumonia can be caused by bacterial/viral infection or fluid in the lungs from aspiration or exposure to cold, damp conditions (especially in infants).

Symptoms include:

- Lethargy
- Difficult, labored or rapid breathing
- Wheezing, crackling or clicking sounds when breathing
- Fever
- Nasal discharge
- May move to top of bedding

Treatment:

- Nuflor along with Baytril
- Keep warm, well hydrated and quiet
- Make sure environment is clean, smoke/dust-free
- Phosphorus (homeopathic treatment)

SNEEZING

- May be due to fluids entering nasal passages when feeding; see "Aspiration" (page 12)
- Check bedding/environment for possible dust or lint that may be getting in nose
- Possible cold/respiration infection – treat with antibiotics (Baytril and Clavamox given together are a possible choice)

DIGESTION/ELIMINATION ISSUES:

ANURIA (Unable to urinate)

Anuria is the lack of urination and is generally caused by dehydration. It is fairly normal for a new squirrel not to urinate for 12 – 24 hours after receipt, but if this persists any longer he may need some help eliminating. It is also possible for urine to be released due to another squirrel sucking, leaving little or no urine when stimulating (see Genital Sucking for remedy). Darker color indicates urine has been kept in the bladder too long, requiring more frequent stimulation. To regain urine flow:

- Provide additional hydration with Ringer's or Electrostat. Try giving 5% of body weight every 2 hours for 2 cycles.
- Gently massage lower abdomen, moving downward toward urethral opening.
- Homeopathic remedy is Plumbum.

BLOATING

Bloating occurs less frequently than diarrhea, but can be severe, even life-threatening. It can be caused by overfeeding, swallowing too much air, or more serious problems, such as parasites. A squirrel is bloated when the stomach feels taut, rather than just full. If your squirrel seems hungry and becomes satiated very quickly, this could be an early indicator of bloating.

- Stimulate the genitals of the squirrel, then place him on his stomach on a heating pad on *Low* or on a warm, wet wash cloth and gently massage his lower back and stomach.
- If he is able to defecate or pass gas, it may relieve the bloat.
- A drop or two of *Infant Simethicone* may be given.
- At the next feeding, try feeding him less and make sure he is not swallowing air.

CONSTIPATION

It is fairly normal for a new squirrel not to defecate for 12 – 24 hours after receipt, but if this persists any longer he may need some help eliminating. Constipation is noticeable if the squirrel is defecating infrequently or not at all, or when the feces are very small, hard, dark, and appear to be difficult or slow to pass. It can result from dehydration, injury, intestinal blockage, going too long without stimulation, or a variety of other causes. Care may include:

- Additional hydration with Ringer's or Electrostat. Try giving 5% of body weight every 2 hours for 2 cycles.
- Homeopathic remedy is Plumbum. (only recommended treatment for pinkies)
- Karo syrup, 2 drops to 10cc formula or 0.2-0.3 cc orally (undiluted) 1 time only.
- Pumpkin puree, 0.4 cc mixed with a bit of water. Usually effective after one treatment, but may repeat after 4 hours. Freeze leftover puree in ice cube tray.
- Pineapple juice, 0.3 cc pineapple juice. Usually effective after one treatment, but may repeat after 4 hours.
- Papaya tablet crushed and mixed with water. One tablet crushed; dissolved in 3 cc water, give 1 cc. Usually effective after one treatment, but may repeat after 4 hours.
- Gently massage lower GI tract, moving across lower abdomen and downward toward anus

DIARRHEA OR SOFT STOOL

Soft stool or diarrhea is the most common problem, and can be caused by any sudden change in diet, too-rich formula, overfeeding, dirty feeding equipment, antibiotic treatment, parasites, bacterial infections and other problems. If left unchecked, diarrhea can cause severe dehydration and can be life-threatening. Symptoms can begin slowly; watch for stool to become sticky, elongated, softer than normal (pudding consistency) (*soft stool*), or watery and free-flowing (*diarrhea*), or for a change in color to brighter yellow, whitish, or greenish.

Dietary steps to try first if stool has begun to get soft/color change (prior to full-on diarrhea):

- Allow longer interval between feedings. (Food may be moving too fast through the digestive system, not allowing time for the proper amount of water to be removed.)
- Consider if it's time to reduce the number of feedings/day, eliminating one and feeding higher volume with subsequent feedings.
- If feeding more than 5%, delay next feeding and reduce subsequent feeding to no more than 5%
- Feed a more dilute formula. (If you are feeding at 2:1 ratio, try 3:1 for several feedings. If no improvement, go to 4:1 for several feedings. Reverse as you see improvement.)
- Omit new solid foods (if weaning) to determine if they are causing the problem.
- Make sure feeding utensils are clean.
- Minimize stress.

Sudden Onset of Diarrhea (when stool has already degraded to full stage diarrhea):

- Skip next feeding entirely.
- Consider hydration at 5% at next feeding, allowing stomach and GI tract to empty. Switch to Ringer's Solution or Electrostat for 1 to 2 feedings.
- Resume feedings at 5%, allow longer interval between feeding.

ANTIBIOTIC USE - Causing diarrhea

- Benebac or Probiacin should be given to maintain intestinal flora when receiving antibiotics. One of these beneficial bacteria should always be given as a preventative, at least once a day, when a squirrel is being administered an antibiotic. Give roughly the same amount as the antibiotic (e.g., if 0.1 cc Clavamox, give 0.1 cc Benebac.) If out of balance due to antibiotic use, stool may appear orange, rust-colored or green.

INTERNAL PARASITES - Causing diarrhea

- Squirrels are subject to illness caused by numerous internal parasites, most often *Coccidia* (the most prevalent), roundworms, hookworms, and *Giardia*. Symptoms of these illnesses may include diarrhea, mucous or blood in stool, loss of appetite, slow growth, mild weight loss to extreme emaciation, general debilitation, and sometimes vomiting. (*Coccidia* may include too soft, voluminous or smelly stool and/or bloody stool.) (*Giardia* may include greenish

color.) Squirrels can be treated for roundworms, tapeworms, *Coccidia*, *Giardia* and other parasites and illnesses with medications.

Possible infection treatments:

- Coccidia or Giardia--stool sample should be tested by a veterinarian. If test is positive, treat with Flagyl for suspected Giardia; Baycox or Albon for Coccidia.
- Worms--treat with Panacur (Worms are not a common problem. Other causes should be addressed prior to treatment for worms.)

RED OR PINK URINE

All rodents tend to process a blood component, uroporphyrin, resulting in reddish colored urine. Fox squirrels are especially known for this. Although bloody urine may be a sign of internal injury or other medical issue, reddish-colored urine may be normal, particularly if it is clear and not cloudy. Consider the squirrel's history in deciding if treatment for internal injury or UTI (urinary tract infection) may be necessary.

VOMITING

Occasionally, a squirrel will come in very thin and vomit after every feeding. This may have several causes, including parasites in the esophagus, injuries, poisoning, or a gastrointestinal disorder, and the squirrel should be seen immediately by your wildlife vet. Poisoning may also cause vomiting.

INFECTIONS/PARASITES:

EAR MITES

If your squirrel is scratching his ears or shaking his head frequently, or if there are any of these other signs: black residue in the ears, redness from inflammation, blood from over-zealous scratching, or edge of ear may look chewed, ear mites should be suspected.

- Use Q-tip to apply mineral oil inside ear, allow to set for several minutes.
- Gently remove any black residue and place used Q-tip in plastic bag with more mineral oil in order to kill the mites.

FLEAS

- Sprinkle small amount of kitten flea powder on a small piece of fleece to put into basket or cage. Place nearby but not on squirrel.
- Capstar is an ingestible method of flea control, but rarely used.

SWOLLEN EYES/EYE INFECTIONS

Occasionally, an infant squirrel's eyes will swell before they open due to an infection. (This is more severe-looking than the "bulgy" look the eyes have at birth to 10 days.) He should be seen and treated by a wildlife vet, who may give him an antibiotic eye ointment. An older baby may also develop an eye infection, for which specialized antibiotic ointments and drops are available.

INJURIES:

ANIMAL BITES

- Spread vasoline or antibiotic gel on hair around wound, then trim hair away. The gel will allow the hair to be whipped away so it doesn't get into the wound.
- Clean injury with antibiotic solution such as Novolsan or Vetericin, being sure to get solution deep into wound. Repeat numerous times per day until wound begins to heal and appears infection free.
- Put antibiotic cream topically on impacted area.
- Begin a course of antibiotics immediately. (Baytril and/or Clavamox are good first choices. If infection does not improve, a combination of antibiotics may be needed.)

MILK SCALD

Missing hair and skin irritation can be the result of formula spilled on the body. Avoid this by making sure to clean the area with warm water after each feeding. If there is further irritation or infection apply triple antibiotic cream.

SEVERE WOUNDS AND BROKEN BONES

Depending on the severity, location, and age of the squirrel, these may or may not be treatable. However, head or facial injuries can usually be treated by a rehabber, and broken extremities can often be set by our assisting vets. Head injuries can include broken cartilage in the nose, jaw injuries, and other head trauma (baby squirrels often seem to land on their heads or noses when they fall).

Head Trauma/Pain

- Metacam can be used to reduce swelling and pain in less serious cases.
- Dexamethasone can be used to reduce swelling and is used for the more severe head injuries and for potential nerve damage such as spinal injuries.
- Antibiotics are necessary for open wounds. Oral medications or topical treatments are available dependent on the type of wound and severity.
- Veterinary care is necessary for broken bones or deep wounds that may need sutures. X-rays may be needed to determine damage, especially if a broken back is suspected.

OTHER ISSUES:

GENITAL SUCKING

Genital sucking is not uncommon among baby squirrels that are caged together, and may result in abrasions, deeper wounds, or infections. Squirrels may also suck on themselves. Although the penis is the most common genital sucked, female genitals and any other body part may also be sucked. Possible causes include teething, stress, lack of adequate fluids, a urinary infection, or inadequate stimulation after feeding. Or it may be that the squirrel's urge to suck is simply not satisfied by his feedings. Increasing the frequency of feedings, feeding smaller amounts than usual, may reduce the sucking, but it may continue despite your best efforts.

- Treat the injured penis topically with a small amount of Neosporin, triple-antibiotic cream, or another antibiotic ointment.
- Put Bitter Apple or "Yuk" ointment around the genital region of the injured squirrel, being careful not to put any directly on any abrasions, to discourage sucking by a nest-mate.
- Penis sucking often results in swelling of the penis or scabbing over of the urethral opening, blocking the flow of urine. Remove the scab by soaking with warm water and repeat frequently until the swelling has gone down, and the free flow of urine is restored.
- Treat any inflammation, showing possible infection, with an oral antibiotic.
- If the condition persists, you may have to separate the squirrels, putting the perpetrator in a nearby cage. Reuniting can be attempted after a few days, but watch carefully for a continuation of the sucking behavior.

LOSS OF FUR

Squirrels go through molts, just as birds do, but patchy fur loss, usually with redness of the skin or actual sores, may also be caused by mange mites, ringworm or another fungus, or a bacterial skin infection. May require advice from vet.

- Ivermectin is used for mange and mites. Place one drop on skin behind the ear or one drop injected into or on a walnut can be fed to the squirrel. Repeat in one week.

METABOLIC BONE DISEASE

The Fox Valley formula contains adequate, balanced amounts of calcium and other nutrients. Once weaning or weaned, though, baby and juvenile squirrels need to be observed for adequate calcium intake. Metabolic Bone Disease, which weakens bones and does not allow them to grow properly, is common when an animal's diet is deficient in calcium, but calcium alone is insufficient. Although there are treatments for MBD, prevention is always preferable.

Your squirrel needs sunlight to produce Vitamin D, which is needed to use calcium efficiently, for good bone growth. Sunlight is an additional and important reason for putting baby squirrels outside where they can get plenty of it during the day as they grow older and begin to wean. If, after weaning, your squirrel cannot be outside for part of the day, you will need to provide a full-spectrum light source during the day. Full-spectrum light bulbs are available at most pet stores, some hardware stores, and on-line, as well as in specialized full-spectrum lamps.

The best solid diet to prevent MBD contains fruits and vegetables (particularly green leafy vegetables) that are high in calcium, phosphorus and other minerals; most nuts are also a good source of calcium. It is also important to provide the correct calcium-to-phosphorus ratio in much of the squirrel's food; this ratio is 1 Ca: 0.5 - 1 Ph. A list of *Calcium-Rich Fruits and Vegetables* is Attachment 3 to this manual. Adding a bird cuttlebone to his cage that he can gnaw on is also recommended to provide an additional source of calcium and other minerals.

If a baby squirrel is not active, climbing and jumping at the appropriate age, and you suspect that he has MBD, you will need to get an x-ray to confirm it.

- Tums tablet may be ground, mixed with water and fed to squirrel.

SCREAMING OR CRYING

Squirrels sometimes let out an ear-piercing scream, or seem to cry continuously. This can be the result of pain, fear, or simply that your hands are too cold. Do not be alarmed at one scream, but if screaming or crying behavior continues something is wrong. Possible causes should be considered and resolved. A veterinarian may need to be consulted.

3. GUIDE TO DEVELOPMENT, FEEDING, HOUSING, AND GENERAL CARE

The following is an approximate guide to stages of development, amounts to feed, and housing and caring for your squirrel. (For additional information on appearance at various ages, see *Squirrel Baby Development* table, Attachment 5 of this manual). The guide is based on the Western Gray Squirrel, but the same principle of weight-based formula feeding, and treatment for illnesses, injuries, and other problems applies to all squirrel species.

The smaller squirrel species develop more quickly, wean themselves sooner, and require a higher percentage of formula each feeding than Western Gray and Fox squirrels. Some other species also require somewhat different housing, solid foods and release procedures than the large tree squirrels, so you will need to work with an experienced rehabber if you get one of these species to rehab. “Mini-manuals” on specific species are also available from your Team Leader or mentor.

Feeding amounts are determined by weight. Although the sections by age, below, show specifics, a good rule of thumb for the amount at each feeding is to feed a maximum amount no greater than 5% of the animal’s body weight in grams, until he is 6 weeks old. After that, you may feed 5.5% up to 6%, as long as the stool remains good. Smaller species, flying squirrels, Douglas squirrels, and chipmunks, require feedings of 6-7% of their body weight from the start.

NEWBORN TO 10 DAYS OLD (Neonates or “Pinkies”)

Note: Caring for newborns is a specialized process and should not be attempted by a novice rehabber. For assistance in raising pinkies, see the *Squirrel Care: Neonates to 2 Weeks* supplemental information under attachments.

Weight: About 10 – 20 grams. Pink all over, with transparent skin, no fur, eyes tightly closed and “bulgy,” ears tightly “glued” to head. Umbilical cord or remainder still attached for first 10 days. Sleeps all the time, except when feeding. On supplemental heat at 85 – 90 degrees F. Chills easily.

Feeding: Intake of formula per feeding: 0.5 – 1 cc every 1.5 – 2 hours, 24 hours a day. Wrap body and keep warm while feeding. Stimulate after each feeding. Urine should be clear or very pale yellow. Before introducing formula, stool will be dark brown and firm. After the first few formula feedings, the stool should be of a dark gold color, firm and round. This is the ideal color and texture until the baby goes on solids. Other colors and textures indicate problems. After stimulation, gently give a

warm "bath" all over by wiping with cotton ball dipped in warm water. Do not dry. This helps prevent drying of the skin.

Housing: Keep in a covered basket with a snap-lid top or a similar container, with air vents, filled with soft, ravel-free bedding and covered with a light towel or blanket. (See housing supplies under *Supplies*, Attachment 1). Place the basket on a heating pad set on Low to maintain an internal temperature of 85 – 90 degrees F. Feel the place in the basket where the squirrel is lying and check your heating pad often to see if it's "toasty warm." Maintain a room temperature of 80 – 85 degrees; a room temperature thermometer can be placed near the basket. A warm, wet washcloth placed over the basket under the covering towel, and/or a small, wet sponge placed in a small, open jar, in a corner of the basket will also help maintain the humidity important to furless, "pinky" squirrels. The baby should feel toasty-warm when you pick him up for feeding.

10 DAYS - 2 WEEKS: INFANT

Weight: About 15 – 35 grams. Pink, very little fur except dark streaks on spine where fur will begin, eyes closed but more slitted, ears a little more defined, whiskers noticeable. Body begins graying more at 2-3 weeks. Umbilical cord should have fallen off completely. Keep baby covered and on supplemental heat (Low) and stimulate to defecate and urinate. Still sleeps most of the time, but can "squirm" around in the basket amazingly, even at this age, so be sure the bedding offers good support. Check basket often, in case squirrel has fallen into a corner, has become cold, too hot, etc.

Feeding: Intake of formula per feeding: 1 cc – 1.5 cc every 2 hours. Once the umbilical cord has fallen off, you can discontinue the "night" (early a.m.) feeding. However, at least one late-night (11:00 – 12:00 pm) and early morning (5:00 – 6:00 am) feeding should be given. An additional feeding (e.g., 3 a.m.) should be continued if the infant is in a weakened condition, not nursing well, or sick. Urine should remain clear or very pale yellow; the stool should be of a dark gold color, firm and round.

Housing: Same as above. Still chills easily, and needs supplemental heat and stimulation.

2 – 3 WEEKS: OLDER INFANT

Weight: About 30 – 60 grams. Becoming bluish-gray and fuzzy, except for undersides; eyes less bulgy and eye slits more obvious; ears more defined. Sleeps most of time. Still on supplemental heat and stimulation.

Feeding: Intake of formula per feeding: 1.5 – 3.0 cc every 2.5 - 3 hours during the day and evening. Continue to give every 3 hours including one mid-morning (approx. 4:00 a.m.) feeding until over 60 grams/3 weeks. Continue to stimulate and use supplemental heat.

Housing: Keep heating pad on Low underneath basket. Clean the basket every few days, leaving some old bedding. Never use chlorine bleach, or anything scented, to wash or dry squirrel bedding; scents irritate their mucous membranes. Hot water and a natural, environmentally safe, unscented detergent, is recommended, but you can use any regular, gentle detergent.

3 – 4 WEEKS: BABY

Weight: About 60 – 90 grams. Skin definitely becoming gray and furry except for undersides; white fur beginning on chest. Reacts to sounds, but sleeps most of time; a little more active after feeding. Eyes still closed, but eye slits are more obvious. Lower incisors (lower front teeth) emerging. You may notice some blood on syringe when feeding, from teething.

Feeding: Intake of formula per feeding: 3 – 4.5 cc every 3.5 hours. Continue to stimulate and use supplemental heat.

Housing: Same as for 2-3 weeks; keep heating pad on Low under basket.

4 – 5 WEEKS: OLDER BABY, EYES CLOSED

Weight: About 90 – 130 grams. Gray fur on topsides; white fur on chest. Eyes beginning to blink (may “wink” at you!); they should open at around 5 weeks. Furry all over by about 5 weeks, but cannot curl tail on own. Reacting more to sounds and starting to explore on own. May begin to make “barking” noise when startled. Lurches around, grabs at things when picked up (“Velcro stage”). Still needs stimulation and supplemental heat. Do not expose squirrel’s face to bright lights or a camera flash for several days after eyes open, as it may damage them.

Feeding: Use 2:1 ratio from this point on. Intake of formula per feeding: 4.5 cc – 6.5 cc every 4 hours.

Housing: Same as above. Place the heating pad, still on Low, under the basket, but under only one end, so he can move around if he gets too warm or cool.

5 – 6 WEEKS: OLDER BABY, EYES FULLY OPEN

Weight: About 130 – 150 grams. Furry all over, eyes and ears open, can curl tail a little. Vision clears after eyes are open a few days. Cutting upper incisors; may try to hold food. May attempt independent elimination, but continue to stimulate him if he'll tolerate it. If his urine is dark or strong-smelling, it has been held in the bladder too long, and the squirrel still needs stimulating at least twice a day.

Feeding: Intake per feeding: 6 cc – 7.5 cc every 4 - 4.5 hours.

Housing: After eyes fully open and he can focus (in about 3 days), move to small cage, approximately 18" to 2 feet W X 2 feet H, of ½" 14-16 gauge welded wire. Add carpeting, indoor-outdoor washable carpet or soft bath mats—without loops—which can be thrown in the washer. You can also add screw-in wooden shelves (available from exotic animal/small pet supply stores or homemade) for feeding and perching. Squirrel babies also enjoy fleece hanging tubes to rest and play in, which are easy to make; some are available from the Squirrel Team or animal supply stores. Construct a "hammock," made from a baby blanket or other small blanket, and hooked or clothes-pinned through the wire; put in several pieces of soft bedding, and hang the hammock very low until your squirrel can climb well, gradually moving it higher, so he can see out better.

Eliminate use of the heating pad during the day unless the squirrel is recovering from an illness. Continue to keep it on at night, still on Low, under only one end of the hammock. Always cover the top of the cage with a large towel or small blanket, since many of the squirrel's predators attack from above, and you want him to feel as secure as possible. Fold sides on top of the cage during the day, and re-drape at night. Provide a hamster-sized hanging water bottle and change water every day; he will soon find it by smell – you don't need to teach him. A large cuttlebone (for calcium) may also be attached inside the cage.

After a few days, provide some thick (1" - 3" or so) branches for climbing (oak, pine, cedar, fir, maple and aspen are good) and cross-branches, with plenty of bark, attached firmly to the inside of the cage with wire or cable ties (leaving wire ends outside the cage and short), and a couple of thicker pieces of wood for him to gnaw

on. Add Timothy or Orchard hay (softer than Alfalfa), about 3-4 inches thick, to completely cover bottom of cage, and change frequently to eliminate wetness and droppings. You can also put in a couple of rocks or a log for climbing and perching. Provide hard things to chew while he is teething, and for the remainder of the time he is caged. A sterile or disinfected dog bone (real, not rawhide) or large piece of deer antler is also good for teething and gnawing. You are beginning to get him used to his wild habitat of trees, grass, shrubs and rocks, rather than cloth and wire.

6 – 7 WEEKS: OLDER BABY TO JUVENILE

Weight: About 150 - 200 grams. Furry all over, including underside of tail and abdomen. Upper teeth more evident. Cutting molars and may start shredding and gnawing things more. Standing and walking more steadily and can sit up with tail curled; may attempt climbing and jumping. May startle when noticing things; may be very vocal and chatter with his siblings and at you. May attempt to hold food. Take particular care from this age on that he doesn't escape from his cage when you are feeding. No supplemental heat, unless recovering from illness.

Feeding: Intake per feeding: 7.5 cc – 10 cc, 4 times per day. Some squirrels like Zoopreem animal chow, and this is the ideal diet to try first. There are a variety of recipes for homemade squirrel chow that some seem to like. Start introducing a few pieces of any of the dry foods at this age. He should begin to nibble on and "shred" it soon, although he won't be actually eating much of it. Some of the smaller species, like ground squirrels and Douglas squirrels, seem to accept the dry block best. We usually end up feeding them both pieces of the dry diet and of regular fruits and vegetables, plus nuts and leafy greens.

If your squirrel refuses to begin nibbling on it, try rubbing the dry diet with an apple, grape, walnut, blueberry, or a little blueberry yogurt. Once he begins to nibble on it, after a few days, begin offering him 1-2 pieces of soft fruit (e.g., avocado, watermelon, banana, peach.) Also provide 2 pieces of the dry food a day. Don't be surprised if your squirrel urinates and defecates on his food; this is the way he stakes a claim to it!

During this stage, if he makes a choking sound or doesn't want his formula, he may have a piece of something stuck in his mouth or throat. If you have to remove anything from your squirrel's mouth, use round-ended tweezers or forceps, under a good, strong light.

Housing: 2' W X 3' H cage with hammock and "natural" furnishings, explained above.

7 - 9 WEEKS - PRE-WEANED JUVENILE

Weight: 200 - 300 grams. Fur is longer and tail is bushy. Squirrel can see and hear well and is more interested in eating solid foods, but still on formula. He is very active, walking, climbing, jumping and running around, and starts sitting up to hold his food. Drinks from water bottle. Teeth are fully developed. He usually will eliminate on his own, but check for droppings to make sure he is doing so, and that they are normal.

Feeding: Intake per feeding: 12.5 - 18 cc, 3 - 4 times a day. You want to start actively weaning your squirrel when he is down to 3X/day of his formula. Make dry diet available at all times. He may begin to eat solid food more readily if you delay his morning feeding a little. In addition to the softer foods provided before, include firmer fruits and vegetables as he begins to chew better (peeled apple, large berries, pear, large grapes, broccoli, carrot, Brussels' sprouts), and large, shelled whole or ½ nuts (walnut, pecan, almonds), or large cracked acorns. (See the *Food List* at the end of *Section 7*.) Select just one or two new foods to start, introducing others gradually.

It may help to hold the food against his front teeth, forcing him (gently) to smell and taste it. Always peel any piece of fruit at this stage and always leave **large** pieces of any food, **so that he cannot get a whole piece in his mouth at one time.** His molars are not developed enough to chew well yet, and he may try to swallow a smaller piece whole, possibly choking (e.g., pieces the size of a small walnut are a good size). Always introduce new solid foods slowly, so his digestive system will get used to them. Provide more natural foods, including cracked nuts, pine and other conifer cones, pines nuts, pumpkin seeds, and branches or logs with moss and lichen on them. **DON'TS:** Do not give any highly-acidic foods, such as oranges, yet. (See *Food List*.) Do not ever give salted nuts. Do not give a baby or juvenile squirrel peanuts or sunflower seeds. A diet restricted to sunflower seeds and peanuts does not contain the essential amino acids necessary for calcium absorption, which can result in Metabolic Bone Disease (see *Section 2*).

Check his incisors (front teeth) for correct positioning; the uppers should be straight down; the lowers are longer, and should curve in slightly; they should meet in a normal bite, the top two overlapping the bottom two evenly. Properly-

positioned teeth are critical to a squirrel's survival, since a squirrel's incisors grow continuously, and gnawing hard things is essential to keep them from growing too long and piercing his soft mouth tissue. If front teeth are lost, and the nerves have not been killed, they will usually grow back. Squirrels also have molars, and move their teeth horizontally to chew.

Housing: Same as above. When the outside temperature is approximately what he is used to indoors, in spring or summer, start acclimating the squirrel to the outdoors by putting his cage outdoors in an area safe from predators and where you can keep an eye on him. Bring him back in when the temperature drops 5 degrees or within a couple of hours, at first. Continue to expose him to temperatures at progressively warmer and colder levels (5 degrees either way), increasing the amount of time outdoors per day, bringing him in before dark. Be sure to keep any household pets away from the squirrel's cage, so that he doesn't become use to them and too trusting of domestic pets. If you don't have a safe place outside, place his cage near an open window for much of the day and be sure to use full-spectrum light bulbs near the cage.

If you can make a small nest box that will fit in the cage, and attach it up high to the wire, he may prefer this to a hammock at this age. Stuff the nest box with blankets, and you may want to add some soft hay. However, using a hammock is also fine throughout his indoor captivity.

He will probably be frightened the first few days outside, and hide in his hammock or nest box. Eventually, he'll begin to get curious about all the sounds and scents he's noticing and come out to see what "the great outdoors" is like. He will gradually grow more comfortable with being outdoors in his cage, and may even "visit" with wild squirrels that come to his cage (but keep an eye out for aggressive wild squirrels and other animals).

Handling: You have been holding your squirrel to feed and clean him, weigh him, possibly give him medicine, and just because squirrel babies are adorable! At this point, you should begin handling him less, primarily only at feeding time (or, of course, if injured or sick). Never let your squirrel run around a room or your house, or climb on you, pretending you're a "tree." This behavior, if allowed, will habituate him far too much to human smells and could even lead to his jumping on other people. Do not allow any of your domestic pets to be around him at any age.

Your squirrel should be encouraged to develop good wild instincts, particularly fear of predators – including dogs, cats, and humans – not to view them as

playthings. The more tame your squirrel becomes, the less developed those instincts will be, and the more danger for him, once he is released. Your squirrel has been so dependent on you for food and care, he thinks of you as his “mother” (strange-looking squirrel that you are!). But your job as a **successful** rehabber, after the initial baby-to-juvenile stages, is to “wild him up,” and eventually break that bond he has developed with you. This can be a little emotionally painful, but that’s the end goal of rehabbing — **doing what is best for the animal.**

It is also a good idea to introduce the squirrel to heavier gloves at this stage, as you may need them as he gets older, and you don't want him frightened by them. Baby squirrels seldom bite their handlers, but their little claws are sharp and can seriously scratch. Start wearing one glove at feeding time, and progress to two when he’s used to it. Sturdy cotton or canvas gardening gloves with reinforced fingers are thick and flexible enough to be useful for handling older baby squirrels up to about 12 weeks. To handle an older juvenile or adult squirrel you will need heavier gloves; soft, flexible leather gloves (such as deerskin), lined with Kevlar, or with Kevlar knit inserts, work very well. Keep your fingers out of the tips of the gloves. Squirrels can bite viciously, with about 1,000 lbs. per square inch pressure from their jaws, and can bite through anything other than Kevlar. A squirrel bite is extremely painful, and you will have to have a tetanus shot and probably take antibiotics if you are bitten, so be very careful to avoid it. (A preliminary tetanus shot is a good idea if you're handling any wildlife.)

9 – 11 WEEKS - WEANING JUVENILE

Weight: 300 - 420 grams. His coat is longer and more dense. At this stage, the squirrel is very active, dashing about, jumping and climbing. He plays, wrestles, pounces on siblings; explores more; hangs upside down from branches, shelves and the cage wire, and starts grooming himself more intensely.

Feeding: Intake per feeding: 18 cc - 25 cc. As the squirrel becomes increasingly fond of solid food, you can begin to reduce his formula feedings to twice daily. His intake will vary with his weight (and personality – some squirrels are real little gluttons!), but, for example (remember the $Wt/2$ formula!), a 300 gram squirrel will take about 15 cc two or three times a day. Some squirrels will wean themselves as early as 9 weeks; others would probably take formula forever, if allowed, but you need to cut them off by the time they are about 3 months old and in the Release

Cage. **Tip:** Adding a little additional water to the formula, making it thinner, may make it less appealing and cause him to seek out solid foods more.

Provide additional fruits, vegetables, berries, nuts, and greens; green, leafy vegetables are extremely important to supply calcium for proper bone and teeth growth, so make sure they are mostly the calcium/phosphorus-rich types. (See *Attachment 3*.) and include more foods from his natural habitat: tree branches with bark, buds, leaves and berries (non-poisonous); acorns and other native nuts; conifer cones; pine nuts; flower bulbs (non-fertilized); shrub branches with berries; foods that may fall from trees (apples, pears, peaches); grapes; grass roots; dandelions; clover; tree branches with moss and lichen (they love this), and anything else you think he might find to eat in your area. Squirrels will eat all of these, plus seeds, some flowers and buds, herbs such as basil, grains, and many other things that grow in nature. He will also enjoy a large clay pot filled with dirt or sand for burying some of his food.

Housing: Cage, 2-3 feet wide X 3 - 4 feet tall, of ½", heavy welded wire, with carpeting, cut to fit the bottom and several inches of hay. Attach his hammock or nest box high up in the cage. Stuff the nest box about half-full with some of his previous bedding, along with soft hay, or green or dry leaves (leave enough room for the squirrel!). Continue to cover the top of the cage with a large towel, small blanket, or branches with leaves etc., to make the squirrel feel as secure from predators as possible. If you have several squirrels, provide a larger, rabbit-sized water bottle, with fresh water daily.

You should create as much of a "wild habitat" in his cage as possible, and get him used to more of his natural food. For "furnishings," use tree branches with bark, some good small logs (with bark), and some largish rocks. If possible, he can by now be permanently outside during the day only in spring, summer, or early fall, in a safe, predator-free place. At night or in winter, if it isn't convenient to keep him in the house, keep him in a protected place, such as a closed garage or shed. Young squirrels still need to be protected from predators; they can be injured or killed by raccoons, weasels, coyotes, and other nocturnal predators, even in sturdy wire cages.

11 - 12 WEEKS - WEANED JUVENILE

Weight: 420 - 450 grams or more. Fur is now longer and tail bushier. Squirrel is very active, climbing, jumping, grooming, playing with siblings. Hangs upside

down from branches and cage. Is becoming much more independent and may dislike any handling.

Feeding: Mother squirrels nurse their young for up to 3-4 months, while gradually supplying them with solid food in the nest and teaching them to leave the nest periodically to climb, jump and gather. Your squirrel should be eating 3-5 pieces of rodent chow or Zoopreem per day or large amounts of fruits, vegetables, nuts and greens, and may still be nursing once to twice a day, (about 20 – 25 cc per feeding). Again, in addition to these, provide many of the types of foods he will find in the wild. Continue to offer walnuts and other nuts, but now in the shell and only slightly cracked, not broken in two, so he can get a tooth in and start learning to chew them open. He might not be able to eat a whole walnut until he's at least 4 months, but he will be able to chew into acorns as soon as he's eating solid food, and will be able to chew into other nuts soon.

He should wean anytime from 9 – 12 weeks, and will make it clear to you by eating less of his formula or refusing it altogether. If he has not self-weaned after 8-10 days on one formula feeding a day, dilute the formula by adding 25% more water each day. Most squirrels will refuse formula as it gets more dilute. You can also try giving him formula in a small bowl, to help wean him, or he may refuse it and choose solid food. The weights for weaning vary widely (i.e., larger tree squirrels at anywhere from 350 to 450 grams; small squirrels such as Douglas squirrels wean much younger, at about 120-140 gs). Some less mature juveniles, or those that are small for their age, may need extra time on formula and may be older when they wean.

Handling: Once your squirrel is weaned, you should STOP handling him, unless it is absolutely necessary, and stop talking to him. Since he is familiar with your hands and the sound of your voice, the less you interact with him the sooner he will complete the process of becoming wild.

12 - 16 WEEKS - GROWING UP, BECOMING WILD

According to field studies, the larger tree squirrels are not capable of nest-building, predator evasion, or foraging for sufficient food on their own, nor do they possess a good sense of direction, until they are about 14 - 16 weeks old. Therefore, it is **imperative** for survival that a squirrel develop these skills prior to release. This indoctrination can be accomplished **only** in a Release Cage for approximately 3.5 - 4 weeks prior to his release to the wild.

A Release Cage should be located near and under trees. The regulation-sized release cage of wood and ½" wire or hardware cloth measures 8' L X 8' W X 8' H (minimum), with a wooden nesting box filled with hay, leaves, or other safe bedding, blanket pieces for extra warmth, and other "furnishings" for feeding and training. Many cages have been built larger. A small "release door," about 3" round or square, that can be fastened shut and is adjacent to an outdoor tree branch, is installed near the top of the cage. Entire 6-8 foot trees and long, thick tree limbs and smaller branches for climbing should be provided. The Release Cage provides predator-proof initial housing that allows juveniles to build the stamina to climb installed near the top of the cage. Entire 6-8 foot trees and long, thick tree limbs and smaller branches for climbing should be provided. The Release Cage provides predator-proof initial housing that allows juveniles to build the stamina to climb vertically, jump greater distances; acclimate more to the outdoors; become accustomed to other wildlife in the area; recognize predators; compete for food (ideally, two or more are housed together); and refresh and clean up their own nests (in the nest box) with materials provided.

Feeding: Your squirrel will continue to be fed in the Release Cage twice a day, with heavy concentration on fruits, greens, other vegetables, nuts and seeds, and natural food he would find in the wild. Squirrels are totally diurnal and "early to bed, early to rise," so it's best to feed them early in the morning and then again mid-to-late-afternoon.

Handling: By now your squirrel should refuse to be touched and should be wary of you, strangers, cats, dogs, and other animals. Even in the release cage, you need to check on your squirrel daily when you feed him to ensure he is active, does not appear to have any illness or injuries, is eating well, and appears to be healthy overall. If your squirrel does appear to be ill or injured (which is rare, once in the release cage, but does occur), you will have to recapture him and treat him. A mesh bird net is essential equipment for capturing an active, older squirrel to care for him.

4. RELEASE

The larger tree squirrels (Gray and Fox squirrels) should not normally be released until the age of 14 - 16 weeks, at well over 450 grams. The smaller squirrel species (Douglas, Flying squirrels, and chipmunks) wean earlier and mature more quickly, and release ages can be as early as 2.5 - 3 months, with weights ranging from 150 up to 300 grams.

Any release site for squirrels should provide a good tree cover and mixture of fresh vegetation, acorns and other nuts from native trees, fruit, berries, fungi and insects, and with an adequate water supply, such as a creek, nearby, or a large watering bowl, freshened daily. The release location must have your species of squirrel living in the area. Since squirrels are diurnal, they should be released very early in the day, to give them a long time to get acclimated to their new surroundings.

By release time, your squirrel's coat should be sleek, his body well-filled-out (even fat), and his tail full and bushy. He should be able to walk, run, jump and climb up, down and horizontally, on the cage and on branches, rapidly and accurately. He should be spending a lot of time playing with his nest mate(s), wrestling, pouncing, exploring, and grooming. He is acting more independent, should be resistant to any handling, and may make efforts to escape from the cage. He will probably be eating large amounts of fruits, vegetables and nuts, along with a wide variety of his natural foods, and be able to drink from a water bottle or to lap. He should also be able to chew into most of his nuts by himself, although you should continue to crack many of the harder ones, like walnuts, enough for him to get a tooth in.

Once the “release door” is opened, the squirrel will eventually find his way out and begin to explore his new surroundings, which is an event well worth watching! Young squirrels put on an act that rivals the Cirque de Soleil, first bouncing off of, and then rapidly climbing, tall trees, jumping from branch to branch, burying nuts, exploring their surroundings and encountering other squirrels and birds. There is nothing as thrilling for a dedicated rehabber as watching a healthy young animal you have nurtured, nursed and rehabilitated running free in the wild, where he belongs. A released squirrel will often return to his cage and nest box to sleep for several days or weeks, until he builds his own nest. You will continue to provide his daily fruits, leafy greens, other vegetables and nuts, plus water, for up to two weeks after release, or as long as he is still returning to the cage.

Tree squirrels are territorial, particularly between species, and resident squirrels will sometimes chase away newcomers. Do not release in an area over-populated by squirrels. Some squirrel species (e.g., Gray Squirrels) are more social, particularly the females, and can live in close proximity, sometimes even sharing nests with siblings or grown offspring, while others (e.g., Douglas and Fox squirrels) are more independent. The larger tree squirrels will usually “wild up” completely at about

9 – 11 months and often go off to find new territories (and mates). Male squirrels, particularly, often leave their first range as they mature, to stake out new territory.

Do not release any squirrels in areas of commercial crops, high traffic, or where their natural habits could cause them to be shot, trapped or poisoned. Although it is illegal to shoot, trap, or poison a tree squirrel in El Dorado County, some people consider them pests and will do so.

5. CONCERNS FOR THE REHABBER

All rodents have sharp, strong teeth, and the handler's tetanus shot should be up to date. Squirrels do not normally transmit diseases to humans (are not *zoonotic*). Rabies is extremely rare in tree squirrels, and it is also very seldom seen in ground squirrels. However, diseases such as Bubonic Plague and other tick or flea-transmitted diseases *may* be carried by any squirrel, particularly adults, although outbreaks of these diseases are tracked regularly by the Department of Public Health and the CDC, and SWR is notified. However, it is important to know where your squirrel was found, and under what circumstances, and to ascertain signs of any severe illness or strange behavior as quickly as possible. Treat any bites from a squirrel seriously, and use a topical antibiotic. If the squirrel shows any signs of a zoonotic illness, the squirrel may need to be tested and the rehabber may need medical consultation/treatment.

6. HANDLING ADULTS (7 MONTHS AND OLDER SQUIRRELS)

An adult squirrel can be extremely hard to handle and can often move quickly and may bite, no matter how badly injured. For those rehabbers who do take adults, it is recommended that a squirrel you have just received be allowed some time to settle down before you examine him. Aside from giving Aconite and Arnica, unless he is bleeding severely or having difficulty breathing, which require immediate treatment, keep him warm (on a heating pad on Low), quiet and in a dark place, with a towel or blanket over the carrier or cage, for 1-2 hours. You will also be able to assess the situation better if you know whether he is willing to eat or drink, and how he moves around in the basket or cage.

Handle him with care, even if badly injured or appearing comatose, because a squirrel can suddenly "revive" and jump, climb and bite, even under these circumstances. An adult squirrel of any species should be handled only with heavy, Kevlar-lined gloves and/or wrapped firmly in a small, heavy blanket. Always wear

protective clothing, such as a heavy shirt or sweatshirt, and keep your fingers out of the tips of your gloves. When handling an adult squirrel, wrap him from front to back in fleece or small blanket ("bundling") and hold him firmly; be sure to control his head by placing a gloved hand over the shoulder area and your other hand at his hindquarters. You can then peel the fleece carefully back from various parts to examine his limbs, body and head, while keeping him firmly restrained. It may be helpful to get assistance from another rehabber when handling an adult.

A wild squirrel usually gives several warning signs when he is anxious and before he attacks, flashing his tail, chattering, and stamping his feet. Another defense mechanism of squirrels is to hold absolutely still and refrain from blinking. Presumably, this is their equivalent of "playing dead," but it should be taken into consideration when giving a physical exam. Squirrels, particularly adults, stress easily, so if there is any sign your squirrel is becoming hysterical while you examine him, wait a few minutes until he settles down, or put him back in the basket, and then try again later. Always approach him quietly and slowly, and handle him gently, but firmly.

When you examine the squirrel, work on a wide, flat surface in a closed room, with plenty of light. Never leave any squirrel unattended, even for a few minutes, unless it is in its cage or carrier with the door securely latched. Again, it may be advisable to have assistance when examining an adult squirrel (and even older wild juveniles), no matter how experienced you are. **NEVER** be reluctant to call someone for help.

7. FOODS FOR WEANING OR WEANED SQUIRRELS

Be sure to include natural foods appropriate to the species of squirrel that you're rehabbing. This list does not include all foods. Research or consult other rehabbers before trying new foods. (Suggestions on "Natural Foods List", based on species of squirrels below)

Nuts and Seeds (also leaves, branches)

Walnuts

Almonds

Chestnuts

Pecans

Hazel nuts (Filberts)

Pine cones and pine nuts

Pumpkin seeds

Squash seeds

Grains (all kinds)

Vegetables

Alfalfa and Clover sprouts

Artichoke leaves – remove to feed*

Broccoli

Brussels sprouts

Carrots

Collard greens

Corn on cob (pieces)

Jicama (ground squirrels in particular)

Kale

Lettuce (except iceberg) – "Spring Mix" variety packaged lettuces/other greens are good

Mushrooms – white or brown

Mustard greens

Spinach

Sweet potato, yams

Squash (all kinds)

Fruits

Apples

Apricots

Avocado (only one piece per day) – no skin, leaves, pit, or stem)

Blackberries

Blueberries

Cherries

Cranberries*

Strawberries*

Grapes

Loquats

Melons (Watermelon or Cantaloupe)

Oranges*

Tangerines/mandarins*

Peaches

Pears

Plums

Prickly pear

Pyracantha berries

Raspberries

Watermelon

Miscellaneous

Zoopreem Primate/ Animal Diet

Homemade Squirrel Chow

Rose petals and hips

Buds and flowers (camellia, rose, manzanita etc. –unfertilized and unsprayed for pests, check edible flower resource before offering)

Cheerios (unsweetened)

* Do not give until your squirrel is 8 weeks old.

8. NATURAL FOODS LIST

Gray & Fox Squirrel

Acorns

Conifer underbark, leaves, cones, nuts, buds

Oak, acorns, underbark, leaves

Maple – underbark

Fungi, lichen, moss

Walnuts, English and Black

Hazelnuts (Filberts)

Spruce

Cedar

Ash

Dogwood -flowers

Manzanita -flowers and berries

Raspberry

Blueberry

Holly (berry)

Hackberry

Elderberry

Pyracantha

Wild grapes

Miner's lettuce

Fruits, Berries

Douglas Squirrel (aka Chickaree, Pine squirrel)

Pine. Fir, Douglas Fir, Cedar, Hazelnut, Ash (underbark, leaves, buds, cones, nuts)

Fungi, lichen, moss

Fruits, berries

Grains (all kinds)

Manzanita -flowers and berries

Miner's lettuce

Seeds and other nuts

Northern Flying Squirrel

Acorns

Fungi, lichen, moss (a lot of these)

Grains (all kinds)

Fruits, berries

Manzanita -flowers and berries

Miner's lettuce

Conifer underbark, leaves, cones, nuts, buds

Seeds and other nuts

Flying insects

California Ground Squirrel/ Chipmunk

Oak

Acorns

Lupine, flower and seed

Oats, wild (veg and seeds)

Seeds and nuts

Manzanita -flowers and berries

Buckeye

Fungi, lichen, moss

Miner's lettuce